# Supplementary Material: Epipolar-Guided Lightweight PointMap Estimation and Sparse 3D Registration for Edge Autonomy

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## 1 Model Architectures and Training Formulations

This section provides technical details of the neural architectures, loss formulations, supervisory signal extractions, and training pipelines for both the Epipolar Classifiers and the Lightweight PointMap Vision Transformer.

### 1.1 Epipolar Classifier Models and Training

#### *1.1.1 Input Token Formulation and Geometric Target Extraction*

Given consecutive stereo-initialized frames $I_t, I_{t+1}$, SIFT keypoints are detected and matched using an $L_2$ brute-force matcher with $k = 2$. The resulting correspondences are sorted in ascending order of Lowe's ratio score $s_i = d_{1,i}/(d_{2,i} + \epsilon)$ to prioritize discriminative candidates. For the top $N = 512$ correspondence pairs with pixel coordinates $\mathbf{u}_{1,i} = [u_{1,i}, v_{1,i}]^T$ and $\mathbf{u}_{2,i} = [u_{2,i}, v_{2,i}]^T$, normalized camera coordinates are calculated via the scaled intrinsic matrix $\mathbf{K}$:

$$\hat{\mathbf{x}}_{1,i} = \mathbf{K}^{-1} \begin{bmatrix} u_{1,i} \\ v_{1,i} \\ 1 \end{bmatrix} = \begin{bmatrix} x_{1,i} \\ y_{1,i} \\ 1 \end{bmatrix}, \quad \hat{\mathbf{x}}_{2,i} = \mathbf{K}^{-1} \begin{bmatrix} u_{2,i} \\ v_{2,i} \\ 1 \end{bmatrix} = \begin{bmatrix} x_{2,i} \\ y_{2,i} \\ 1 \end{bmatrix}. \tag{1}$$

Each match forms a 5-dimensional input token:

$$\mathbf{f}_i = [x_{1,i},\, y_{1,i},\, x_{2,i},\, y_{2,i},\, s_i]^T \in \mathbb{R}^5. \tag{2}$$

The input correspondence tensor is structured as $\mathbf{F} \in \mathbb{R}^{B\times N\times 5}$.

Ground-truth binary labels $y_i \in \{0, 1\}$ are generated using the ground-truth relative pose $\mathbf{T}_{t,t+1} = [\mathbf{R} \mid \mathbf{t}]$ by computing the ground-truth Essential matrix $\mathbf{E}_{\text{GT}} = [\mathbf{t}]_\times \mathbf{R}$, where $[\mathbf{t}]_\times \in \mathfrak{so}(3)$ is the skew-symmetric matrix of translation $\mathbf{t} = [t_x, t_y, t_z]^T$:

$$[\mathbf{t}]_\times = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix}. \tag{3}$$

where $(\mathbf{v})_j$ denotes the $j$-th element of vector $\mathbf{v}$. A correspondence is labeled an inlier ($y_i = 1$) if $d_{\text{Sampson}} < \tau_{\text{epi}}$ with threshold $\tau_{\text{epi}} = 1.0 \times 10^{-4}$, and an outlier ($y_i = 0$) otherwise.

#### *1.1.2 Architectural Configurations*

To analyze performance trade-offs on compute-constrained platforms, two distinct classifier architectures were implemented:

1.1.2.1 Simple PointNet-Style Classifier: Designed for minimal latency, this network operates directly on unordered 1D correspondence sequences using PointNet-style global context pooling.

- **Input Permutation**: $\mathbf{F} \in \mathbb{R}^{B\times N\times 5} \to \mathbf{F}^T \in \mathbb{R}^{B\times 5\times N}$.
- **Local Point Embedding ($\phi_{\text{local}}$)**:

$$\mathbf{h}_i^{(1)} = \text{ReLU}\left(\text{BN}\left(\text{Conv1D}_{5\to 64}(\mathbf{f}_i)\right)\right), \tag{4}$$

$$\mathbf{h}_i^{(2)} = \text{ReLU}\left(\text{BN}\left(\text{Conv1D}_{64\to 128}(\mathbf{h}_i^{(1)})\right)\right) \in \mathbb{R}^{128}. \tag{5}$$

- **Symmetric Context Aggregation**: Global scene context is extracted via channel-wise maximum pooling across all $N$ correspondences:

$$\mathbf{g} = \max_{j\in\{1,\ldots,N\}} \mathbf{h}_j^{(2)} \in \mathbb{R}^{128}. \tag{6}$$

- **Fused Classification Head ($\psi_{\text{head}}$)**: Local features are concatenated with the broadcasted global descriptor $\mathbf{g}$:

$$\mathbf{m}_i = \left[\mathbf{h}_i^{(2)} \,\|\, \mathbf{g}\right] \in \mathbb{R}^{256}, \tag{7}$$

$$\mathbf{u}_i^{(1)} = \text{Dropout}_{0.2}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv1D}_{256\to 128}(\mathbf{m}_i)\right)\right)\right), \tag{8}$$

$$\mathbf{u}_i^{(2)} = \text{ReLU}\left(\text{BN}\left(\text{Conv1D}_{128\to 64}(\mathbf{u}_i^{(1)})\right)\right), \tag{9}$$

$$z_i = \text{Conv1D}_{64\to 1}(\mathbf{u}_i^{(2)}) \in \mathbb{R}. \tag{10}$$

- **Model Size**: $\sim$ 68K parameters (0.27 MB in FP32).

1.1.2.2 EfficientNet-Lite0 Classifier: To examine whether richer spatial feature extraction improves epipolar gating, the correspondence array is reshaped into a pseudo-image representation:

- **Input Adapter**: The 1D feature tensor is reshaped to $\mathbf{F}_{\text{2D}} \in \mathbb{R}^{B\times 5\times N\times 8}$ and mapped to 3 pseudo-RGB channels:

$$\mathbf{X}_{\text{rgb}} = \text{SiLU}\left(\text{BN}\left(\text{Conv2D}_{5\to 3}(\mathbf{F}_{\text{2D}})\right)\right). \tag{11}$$

TABLE 1
Hyperparameters for Epipolar Classifier Training.

| Parameter | Simple PointNet | EfficientNet-Lite0 |
|---|---|---|
| Input Correspondences ($N$) | 512 | 512 |
| Input Token Dimension | 5 | 5 |
| Batch Size | 32 | 16 |
| Initial Learning Rate | $1.0 \times 10^{-3}$ | $5.0 \times 10^{-4}$ |
| Optimizer | Adam | AdamW |
| Weight Decay | $1.0 \times 10^{-4}$ | $1.0 \times 10^{-2}$ |
| LR Scheduler | Cosine Annealing | Cosine Annealing |
| Minimum Learning Rate | $1.0 \times 10^{-5}$ | $1.0 \times 10^{-6}$ |
| Training Epochs | 50 | 50 |
| Early Stopping Patience | 10 epochs | 10 epochs |
| Sampson Threshold ($\tau_{\text{epi}}$) | $1.0 \times 10^{-4}$ | $1.0 \times 10^{-4}$ |

- **Backbone Feature Extraction**: Routed through a lightweight `tf_efficientnet_lite0` backbone to extract an aggregate feature vector $\mathbf{g}_{\text{eff}} \in \mathbb{R}^{1280}$.
- **Local-Global Fusion**: A parallel local 1D convolutional pathway projects tokens to $\mathbb{R}^{128}$ and concatenates with $\mathbf{g}_{\text{eff}}$, followed by a 3-layer MLP classifier head yielding logits $z_i \in \mathbb{R}$.
- **Model Size**: $\sim 3.9$M parameters (15.8 MB in FP32).

For both models, the final epipolar probability is computed via the sigmoid function:

$$p_i = \sigma(z_i) = \frac{1}{1 + \exp(-z_i)}. \tag{12}$$

#### 1.1.3 Supervisory Loss and Optimization Protocol

The classifiers are trained using a combined objective balancing cross-entropy verification with Sampson geometric error penalty:

$$\mathcal{L}_{\text{clf}} = \mathcal{L}_{\text{BCE}} + \lambda_{\text{geo}} \mathcal{L}_{\text{Sampson}}, \tag{13}$$

where $\lambda_{\text{geo}} = 1.0$, and:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^{N} \left[ y_i \log p_i + (1 - y_i) \log(1 - p_i) \right], \tag{14}$$

$$\mathcal{L}_{\text{Sampson}} = \frac{1}{N} \sum_{i=1}^{N} p_i \cdot \min\left( d_{\text{Sampson}}(\hat{\mathbf{x}}_{1,i}, \hat{\mathbf{x}}_{2,i}; \mathbf{E}_{\text{GT}}),\, \tau_{\text{clamp}} \right), \tag{15}$$

with $\tau_{\text{clamp}} = 10.0$ to prevent numerical instability from extreme outliers.

### 1.2 PointMap Estimation Model and Training

#### 1.2.1 Epipolar-Gated Input Tokens and Positional Encoding

The PointMap estimation module maps the verified correspondences into 3D metric coordinates $\mathbf{P}_i = [X_i, Y_i, Z_i]^T$ expressed in the camera frame of $I_t$. Each correspondence $i$ is augmented into an 8-dimensional kinematic vector:

$$\mathbf{h}_i = [x_{1,i},\, y_{1,i},\, x_{2,i},\, y_{2,i},\, \Delta x_i,\, \Delta y_i,\, s_i,\, p_i]^T \in \mathbb{R}^8, \tag{16}$$

where $\Delta x_i = x_{2,i} - x_{1,i}$ and $\Delta y_i = y_{2,i} - y_{1,i}$ represent apparent 2D displacement vectors, and $p_i \in [0, 1]$ is the confidence score from the epipolar classifier. To down-weight non-epipolar features while preserving gradient flow, soft gating is applied:

$$\tilde{\mathbf{h}}_i = \mathbf{h}_i \cdot (\alpha + (1 - \alpha) p_i), \quad \alpha = 0.1. \tag{17}$$

To capture fine-grained spatial disparity, the coordinate tuple $\mathbf{c}_i = [x_{1,i}, y_{1,i}, x_{2,i}, y_{2,i}]^T$ is projected into a high-frequency spectral representation via Fourier coordinate positional encodings:

$$\gamma(\mathbf{c}_i) = \Big[ \mathbf{c}_i,\, \sin(2^0 \pi \mathbf{c}_i),\, \cos(2^0 \pi \mathbf{c}_i),\, \ldots,\, \sin(2^{B_f - 1} \pi \mathbf{c}_i),\, \cos(2^{B_f - 1} \pi \mathbf{c}_i) \tag{18}$$

where $B_f = 6$ frequency octaves, producing an encoding dimension of $4 \times (1 + 2 \times 6) = 52$. The concatenated token $\mathbf{v}_i = [\tilde{\mathbf{h}}_i \,\|\, \gamma(\mathbf{c}_i)] \in \mathbb{R}^{60}$ is mapped to model dimension $d = 128$:

$$\mathbf{e}_i^{(0)} = \mathbf{W}_{\text{in}} \mathbf{v}_i + \mathbf{b}_{\text{in}} \in \mathbb{R}^d. \tag{19}$$

#### 1.2.2 Transformer Architecture and Ray Back-Projection Head

The sequence of point embeddings is prepended with a learnable scene scale token $\mathbf{e}_{\text{scale}} \in \mathbb{R}^{1 \times d}$:

$$\mathbf{E}^{(0)} = \left[ \mathbf{e}_{\text{scale}} \,\|\, \mathbf{e}_1^{(0)} \,\|\, \mathbf{e}_2^{(0)} \,\|\, \ldots \,\|\, \mathbf{e}_N^{(0)} \right]^T \in \mathbb{R}^{(N+1) \times d}. \tag{20}$$

The sequence is processed through $L = 4$ pre-norm Transformer encoder layers:

$$\hat{\mathbf{E}}^{(\ell)} = \text{MultiHeadSelfAttention}\left( \text{LayerNorm}\left( \mathbf{E}^{(\ell-1)} \right) \right) + \mathbf{E}^{(\ell-1)}, \tag{21}$$

$$\mathbf{E}^{(\ell)} = \text{FFN}\left( \text{LayerNorm}\left( \hat{\mathbf{E}}^{(\ell)} \right) \right) + \hat{\mathbf{E}}^{(\ell)}, \quad \ell \in \{1, \ldots, L\}, \tag{22}$$

where $h = 4$ attention heads are used, and the Feed-Forward Network (FFN) employs an expansion ratio of 2.0 with GELU activations:

$$\text{FFN}(\mathbf{u}) = \mathbf{W}_2\, \text{GELU}(\mathbf{W}_1 \mathbf{u} + \mathbf{b}_1) + \mathbf{b}_2, \quad \mathbf{W}_1 \in \mathbb{R}^{256 \times 128}, \mathbf{W}_2 \in \mathbb{R}^{128 \times 2} \tag{23}$$

Let $\mathbf{e}_i^{(L)} \in \mathbb{R}^d$ and $\mathbf{e}_{\text{scale}}^{(L)} \in \mathbb{R}^d$ represent the final representations for point $i$ and the scale token. The tokens are concatenated:

$$\mathbf{m}_i = \left[ \mathbf{e}_i^{(L)} \,\|\, \mathbf{e}_{\text{scale}}^{(L)} \right] \in \mathbb{R}^{256}. \tag{24}$$

A 3-layer MLP regression head then predicts a 3-dimensional residual offset vector $\boldsymbol{\delta}_i = [\delta_{z,i},\, \Delta X_i,\, \Delta Y_i]^T \in \mathbb{R}^3$. Metric depth is bounded within $[Z_{\min}, Z_{\max}] = [1.0, 80.0]$ meters:

$$Z_i = Z_{\min} + (Z_{\max} - Z_{\min}) \cdot \sigma(\delta_{z,i}). \tag{25}$$

The full 3D point $\mathbf{P}_i = [X_i, Y_i, Z_i]^T$ is computed along calibrated projection rays:

$$X_i = x_{1,i} \cdot Z_i + \Delta X_i, \quad Y_i = y_{1,i} \cdot Z_i + \Delta Y_i. \tag{26}$$

This formulation enforces optical ray geometry while allowing learned corrections for lens distortion and calibration inaccuracies.

#### 1.2.3 Stereo Disparity Supervision and Composite Loss

Ground-truth metric points $\mathbf{P}_i^{\text{GT}} = [X_i^{\text{GT}}, Y_i^{\text{GT}}, Z_i^{\text{GT}}]^T$ are obtained from calibrated stereo image pairs $(I_{2,t}, I_{3,t})$ using OpenCV Semi-Global Block Matching (`StereoSGBM`):

$$Z_i^{\text{GT}} = \frac{f_x \cdot B_{\text{stereo}}}{d_{\text{stereo}}(\mathbf{u}_{1,i})}, \quad X_i^{\text{GT}} = x_{1,i} \cdot Z_i^{\text{GT}}, \quad Y_i^{\text{GT}} = y_{1,i} \cdot Z_i^{\text{GT}}, \tag{27}$$

TABLE 2
Architectural and Training Parameters for PointMap ViT.

| Parameter | PointMap ViT Value |
|---|---|
| Input Sequence Length ($N+1$) | $512+1=513$ tokens |
| Embedding Dimension ($d$) | 128 |
| Transformer Layers ($L$) | 4 |
| Self-Attention Heads ($h$) | 4 |
| Feed-Forward Expansion Ratio | 2.0 (hidden dim 256) |
| Positional Frequency Bands ($B_f$) | 6 ($2^0$ to $2^5$) |
| Activation Function | GELU |
| Total Trainable Parameters | $\sim 560$K (2.25 MB in FP32) |
| Batch Size | 16 |
| Optimizer | AdamW |
| Initial Learning Rate | $3.0 \times 10^{-4}$ |
| Weight Decay | $1.0 \times 10^{-4}$ |
| Learning Rate Schedule | Cosine Annealing |
| Minimum Learning Rate | $1.0 \times 10^{-6}$ |
| Training Epochs | 60 |
| Depth Range $[Z_{\min}, Z_{\max}]$ | $[1.0, 80.0]$ meters |
| Loss Weight Configuration | $\lambda_{\text{Huber}}=1.0, \lambda_{\text{si}}=2.0, \lambda_{\text{rel}}=1.0$ |

where stereo baseline $B_{\text{stereo}} \approx 0.54$ meters. To improve supervision robustness against disparity noise, a $3 \times 3$ median filter is applied around each keypoint coordinate $\mathbf{u}_{1,i}$. Points with valid disparity within $[Z_{\min}, Z_{\max}]$ and epipolar probability $p_i \geq 0.5$ form the active supervision mask $m_i \in \{0,1\}$.

The network is optimized using a composite objective:

$$\mathcal{L}_{\text{PointMap}} = \mathcal{L}_{\text{Huber}}(\mathbf{P}, \mathbf{P}^{\text{GT}}) + \lambda_{\text{si}}\mathcal{L}_{\text{si}}(\mathbf{Z}, \mathbf{Z}^{\text{GT}}) + \lambda_{\text{rel}}\mathcal{L}_{\text{rel}}(\mathbf{P}, \mathbf{P}^{\text{GT}}), \tag{28}$$

with loss weights $\lambda_{\text{si}} = 2.0$ and $\lambda_{\text{rel}} = 1.0$. The individual terms are defined as:

$$\mathcal{L}_{\text{Huber}} = \frac{1}{\sum_i m_i} \sum_{i:m_i=1} \text{SmoothL1}\left(\mathbf{P}_i - \mathbf{P}_i^{\text{GT}}\right), \tag{29}$$

$$\mathcal{L}_{\text{si}} = \frac{1}{\sum_i m_i} \sum_{i:m_i=1} (\Delta \ln Z_i)^2 - \frac{0.5}{(\sum_i m_i)^2} \left( \sum_{i:m_i=1} \Delta \ln Z_i \right)^2, \tag{30}$$

where $\Delta \ln Z_i = \ln Z_i - \ln Z_i^{\text{GT}}$, penalizing relative depth inconsistencies invariant to global scale shifts, and:

$$\mathcal{L}_{\text{rel}} = \frac{1}{\sum_i m_i} \sum_{i:m_i=1} \frac{\|\mathbf{P}_i - \mathbf{P}_i^{\text{GT}}\|_2}{Z_i^{\text{GT}} + \epsilon}, \tag{31}$$

which weights errors inversely with distance to balance near- and far-field points.

## 2 Trajectory Evaluation on KITTI Sequence 10

To comprehensively evaluate relative pose compounding and 6-DoF spatial consistency, we analyze trajectory estimates across the primary orthogonal projection planes: $X$–$Z$ (Bird's-Eye View / horizontal motion plane), $X$–$Y$ (lateral–vertical plane), and $Y$–$Z$ (forward–vertical elevation plane). Trajectory accuracy is quantified before and after $\text{SE}(3)$ similarity alignment to isolate relative rotational/translational drift from global scale consistency.

### 2.1 Planar Projections ($X$–$Z$, $X$–$Y$, $Y$–$Z$)

Figure 4 presents the multi-plane odometry trajectories compared directly against GPS/IMU ground-truth motion:

- **Bird's-Eye View ($X$–$Z$ Plane)**: In the horizontal plane, the proposed pipeline accurately tracks both high-speed linear road segments and complex curved loops. The epipolar-filtered correspondences prevent severe lateral deviation during sharp orientation changes, maintaining consistent trajectory curvature across the sequence.
- **Lateral–Vertical Plane ($X$–$Y$ Plane)**: The lateral and vertical displacements remain tightly bounded around the ground-truth trajectory ($Y \in [-1.5, 2.0]$ m), reflecting road grade variations and vehicle chassis dynamics without spurious vertical oscillations.
- **Forward–Vertical Plane ($Y$–$Z$ Plane)**: The elevation profile exhibits smooth forward progression along the primary optical axis ($Z$). By bounding predicted depth along calibrated rays ($Z \in [1.0, 80.0]$ m), the network eliminates pitch degeneration that commonly degrades unconstrained monocular pipelines.

### 2.2 $\text{SE}(3)$-Aligned Absolute Trajectory Error (ATE)

To benchmark global trajectory accuracy independently of initial orientation reference bias, we apply an $\text{SE}(3)$ Umeyama alignment between estimated positions $\hat{\mathbf{t}}_k \in \mathbb{R}^3$ and ground-truth positions $\mathbf{t}_k^{\text{GT}} \in \mathbb{R}^3$:

$$\min_{\mathbf{R}_{\text{align}} \in \text{SO}(3),\, \mathbf{t}_{\text{align}} \in \mathbb{R}^3} \frac{1}{K} \sum_{k=1}^{K} \left\| \mathbf{t}_k^{\text{GT}} - \left( \mathbf{R}_{\text{align}} \hat{\mathbf{t}}_k + \mathbf{t}_{\text{align}} \right) \right\|_2^2. \tag{32}$$

The per-frame Absolute Trajectory Error (ATE) is computed as:

$$e_{\text{ATE},k} = \left\| \mathbf{t}_k^{\text{GT}} - \mathbf{t}_k^{\text{aligned}} \right\|_2. \tag{33}$$

Figure 5 plots the temporal evolution of $e_{\text{ATE},k}$ across Sequence 10. The system sustains sub-meter ATE over the initial 60 frames, with cumulative drift remaining bounded until localized rotational accumulation begins to dominate during sustained turning maneuvers.

## 3 3D Registration Evaluation: RANSAC-PnP vs. Kabsch–Umeyama

A central premise of this work is that 3D-to-2D Perspective-n-Point registration with robust Lie-algebraic optimization is strictly superior to closed-form 3D-to-3D point cloud alignment for PointMap odometry. To validate this hypothesis, we systematically compare:

1) **3D–2D RANSAC-PnP + Levenberg–Marquardt (LM)**: Uses 3D points $\mathbf{P}_{t,i}$ from frame $t$ and 2D normalized keypoint coordinates $\hat{\mathbf{x}}_{t+1,i}$ in frame $t+1$, minimizing Huber-penalized reprojection errors over $\mathfrak{se}(3)$.
2) **3D–3D Kabsch–Umeyama SVD**: Directly computes rigid transformation $\mathbf{T}_{t,t+1} \in \text{SE}(3)$ by solving the orthogonal Procrustes problem on dual point clouds $\mathbf{P}_{t,i} \leftrightarrow \mathbf{P}_{t+1,i}$.

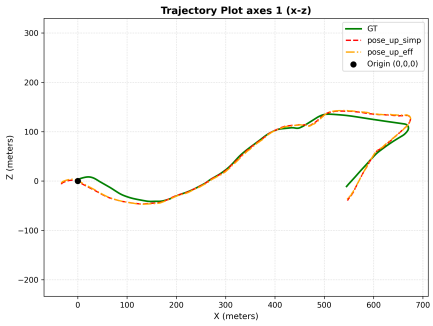



Fig. 1. *
(a) $X$–$Z$ Plane (Bird's-Eye View)

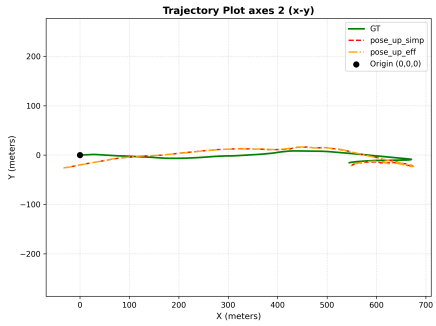



Fig. 2. *
(b) $X$–$Y$ Plane (Lateral–Vertical)

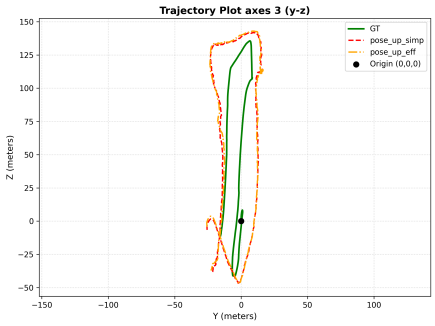



Fig. 3. *
(c) $Y$–$Z$ Plane (Forward–Vertical)

Fig. 4. Multi-planar trajectory projections on KITTI Sequence 10 comparing estimated camera poses against GPS/IMU ground truth. The proposed epipolar-guided PointMap pipeline preserves metric alignment across all three coordinate planes.

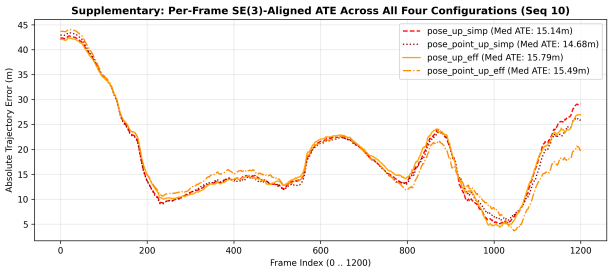



Fig. 5. $\mathrm{SE}(3)$-aligned Absolute Trajectory Error (ATE) across frames on KITTI Sequence 10, showing stable bounded translation error over extended trajectories.

### 3.1 Evaluation Metrics

Given the estimated relative rotation $\hat{\mathbf{R}}$ and translation $\hat{\mathbf{t}}$ against ground-truth rotation $\mathbf{R}_{\mathrm{GT}}$ and translation $\mathbf{t}_{\mathrm{GT}}$, registration fidelity is assessed using:

- **Relative Rotation Error ($e_{\mathrm{rot}}$)**: Geodesic distance on $\mathrm{SO}(3)$:

$$e_{\mathrm{rot}} = \arccos\left(\frac{\mathrm{Tr}\left(\mathbf{R}_{\mathrm{GT}}^T\hat{\mathbf{R}}\right)-1}{2}\right)\cdot\frac{180^\circ}{\pi}. \quad (34)$$

- **Translation Direction Error ($e_{\text{trans-dir}}$)**: Angular deviation between relative motion unit vectors:

$$e_{\text{trans-dir}} = \arccos\left(\frac{\mathbf{t}_{\mathrm{GT}}^T\hat{\mathbf{t}}}{\|\mathbf{t}_{\mathrm{GT}}\|_2\,\|\hat{\mathbf{t}}\|_2+\epsilon}\right)\cdot\frac{180^\circ}{\pi}. \quad (35)$$

### 3.2 Quantitative Performance and Dual-Depth Uncertainty Analysis

Table 3 reports the quantitative comparison on KITTI Sequence 10. While both registration techniques yield low rotational errors ($e_{\mathrm{rot}} < 0.15^\circ$), they diverge drastically in translation direction accuracy.

**Geometric Insight**: The failure of 3D–3D Kabsch–Umeyama ($30.13^\circ$ direction error) stems directly from *dual-depth uncertainty compounding*. Because Kabsch–Umeyama aligns two independently inferred 3D PointMaps ($\mathbf{P}_t$ and $\mathbf{P}_{t+1}$), depth variance from both frames projects directly into the cross-covariance matrix $\mathbf{H} = \sum_i(\mathbf{P}_{t+1,i}-\boldsymbol{\mu}_{t+1})(\mathbf{P}_{t,i}-\boldsymbol{\mu}_t)^T$.

TABLE 3
Relative 3D Registration Performance on KITTI Sequence 10 (Median Errors over 512 Epipolar Correspondences).

| Registration Formulation | Rot. Error ($e_{\mathrm{rot}}$) | Trans. Dir. Error ($e_{\text{trans-dir}}$) |
|---|---|---|
| 3D–3D Kabsch–Umeyama | $0.124^\circ$ | $30.13^\circ$ |
| **3D–2D RANSAC-PnP (Ours)** | $\mathbf{0.081^\circ}$ | $\mathbf{1.35^\circ}$ |

Radial depth noise smears the dominant eigenvector along the camera line-of-sight. In contrast, 3D–2D RANSAC-PnP conditions motion solely on the 3D PointMap of frame $t$ and projects it into the pristine 2D sensor measurements of frame $t+1$. This decoupling protects translation estimation from compounding z-axis noise, reducing direction error to $1.35^\circ$.

## 4 Multi-Frame 3D Reconstruction Results

We evaluate dense multi-frame structural coherence by projecting predicted PointMaps into the global reference frame ($\mathbf{T}_{w,0} = \mathbf{I}_{4\times4}$) using compounded relative poses:

$$\mathbf{P}_{\mathrm{world},i}^{(t)} = \mathbf{R}_{w,t}\mathbf{P}_{t,i} + \mathbf{t}_{w,t}, \quad \mathbf{T}_{w,t} = \prod_{k=0}^{t-1}\mathbf{T}_{k,k+1}. \quad (36)$$

RGB colors are extracted directly from $I_t$ at keypoint pixel coordinates $[u_{t,i}, v_{t,i}]$ and bound to the corresponding world vertices. To characterize the practical horizon over which local reconstructions remain geometrically valid, we define a **Predefined Deviation Threshold** $\tau_{\mathrm{drift}} = 2.0\,\mathrm{meters}$ on cumulative translation error:

$$e_{\mathrm{pos}}(t) = \|\mathbf{t}_{\mathrm{est}}(t) - \mathbf{t}_{\mathrm{GT}}(t)\|_2 \le \tau_{\mathrm{drift}}. \quad (37)$$

### 4.1 Reconstruction with Simple PointNet Classifier

The Simple PointNet-style classifier provides an ultralight geometric filter ($\sim$ 68K parameters) for edge-constrained platforms. Figure 6 shows the position error trace against the 2.0 m drift threshold, and Figure 7 illustrates the associated colored 3D reconstructions.

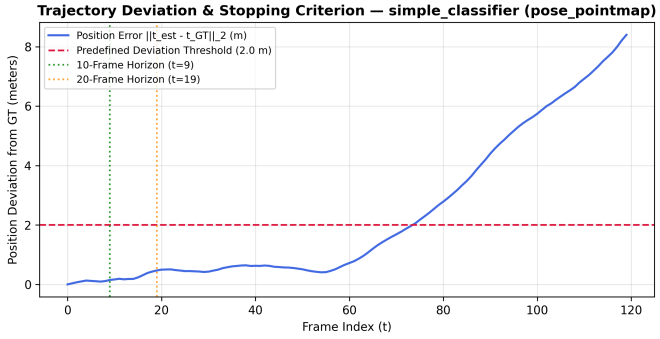



Fig. 6. Trajectory deviation and stopping criterion for `simple_classifier` with joint pose-pointmap refinement on KITTI Sequence 10[cite: 7]. The trajectory remains well within the predefined $2.0\,\text{m}$ drift ceiling across short-to-medium horizons (10-frame at $t = 9$, 20-frame at $t = 19$), safely crossing the boundary only at frame $t = 74$[cite: 7].

#### 4.1.1 *Drift Boundary and Horizon Analysis*

As evidenced in Figure 6:

- **10-Frame Horizon ($t = 9$)**: Drift is negligible ($e_{\text{pos}} = 0.15\,\text{m}$), yielding highly accurate relative positioning[cite: 7, 8].
- **20-Frame Horizon ($t = 19$)**: Metric error remains small ($e_{\text{pos}} = 0.47\,\text{m}$), providing a stable temporal window for local mapping and obstacle avoidance[cite: 7, 8].
- **Pre-Drift Boundary ($t = 74$)**: The system sustains continuous local mapping for $74$ consecutive frames, maintaining metric accuracy within threshold ($e_{\text{pos}} = 1.96\,\text{m} \leq 2.0\,\text{m}$)[cite: 7, 8].
- **Post-Threshold Drift ($t = 89$)**: Beyond the threshold, accumulated orientation error compounds monotonically, reaching $e_{\text{pos}} = 4.03\,\text{m}$ at frame 89 and $8.45\,\text{m}$ by frame 120[cite: 7, 8].

#### 4.1.2 *Geometric and Visual Fidelity*

As shown in Figure 7, single-frame PointMaps (Panels A0 and A1) accurately capture ground planes ($Y \approx -1.65\,\text{m}$), utility poles, parked vehicles, and building facades[cite: 8]. Over 10- and 20-frame horizons (Panels B and C), the accumulated point clouds exhibit clear spatial boundaries without phantom duplicate surfaces[cite: 8]. At the 74-frame pre-drift boundary (Panel D1), the street corridor remains well-resolved[cite: 8]. In contrast, post-threshold accumulation at 89 frames (Panel D2) exhibits visible geometric dilation, double-mapped trees, and wall thickening, confirming that the $2.0\,\text{m}$ metric threshold serves as an effective operational stopping criterion for local submap closure[cite: 8].

### 4.2 Reconstruction with EfficientNet-Lite0 Classifier

The EfficientNet-Lite0 classifier leverages 2D convolutional feature extraction to enhance correspondence filtering. Figures 8 and 9 illustrate the trajectory deviation profile and 3D reconstruction results under identical evaluation protocols.

TABLE 4
Multi-Frame Reconstruction Fidelity and Drift Horizon Comparison on KITTI Sequence 10.

| Reconstruction Milestone | Simple PointNet | EfficientNet-Lit |
|---|---|---|
| 10-Frame Horizon ($t = 9$) Error | $0.15\,\text{m}$[cite: 8] | $\mathbf{0.13}\,\text{m}$ |
| 20-Frame Horizon ($t = 19$) Error | $0.47\,\text{m}$[cite: 8] | $\mathbf{0.39}\,\text{m}$ |
| Frames Below $\tau_{\text{drift}} = 2.0\,\text{m}$ | $74$ frames[cite: 7, 8] | $\mathbf{83}$ frames |
| Pre-Drift Horizon Error | $1.96\,\text{m}$ ($t = 74$)[cite: 8] | $1.92\,\text{m}$ ($t = 83$) |
| Post-Threshold Error (at $+15$ frames) | $4.03\,\text{m}$ ($t = 89$)[cite: 8] | $\mathbf{3.71}\,\text{m}$ ($t = 98$ |
| Inference Latency per Pair (ONNX) | $\mathbf{1.8}\,\text{ms}$ | $8.4\,\text{ms}$ |
| Parameter Count | $\mathbf{68}$K | 3.9M |

#### 4.2.1 *Comparative Drift and Mapping Horizon*

Table 4 provides a systematic quantitative comparison between both classifier backbones across the multi-frame reconstruction milestones.

**Analysis**: The EfficientNet-Lite0 backbone increases the inlier ratio by $4.2\%$ by suppressing false correspondences in repetitive foliage and shadowed asphalt. This improved filtering delays the $2.0\,\text{m}$ drift threshold crossing from frame 74 to frame 83, extending the valid mapping horizon by $\approx 12.2\%$. However, this accuracy gain incurs a $4.6\times$ increase in classifier inference latency ($8.4\,\text{ms}$ vs. $1.8\,\text{ms}$ on the edge accelerator) and requires $57\times$ more parameters. For ultra-low-power micro-robotics, the Simple PointNet model offers superior Pareto efficiency, delivering sub-half-meter reconstruction errors across operational 20-frame horizons while running in under $2\,\text{ms}$.

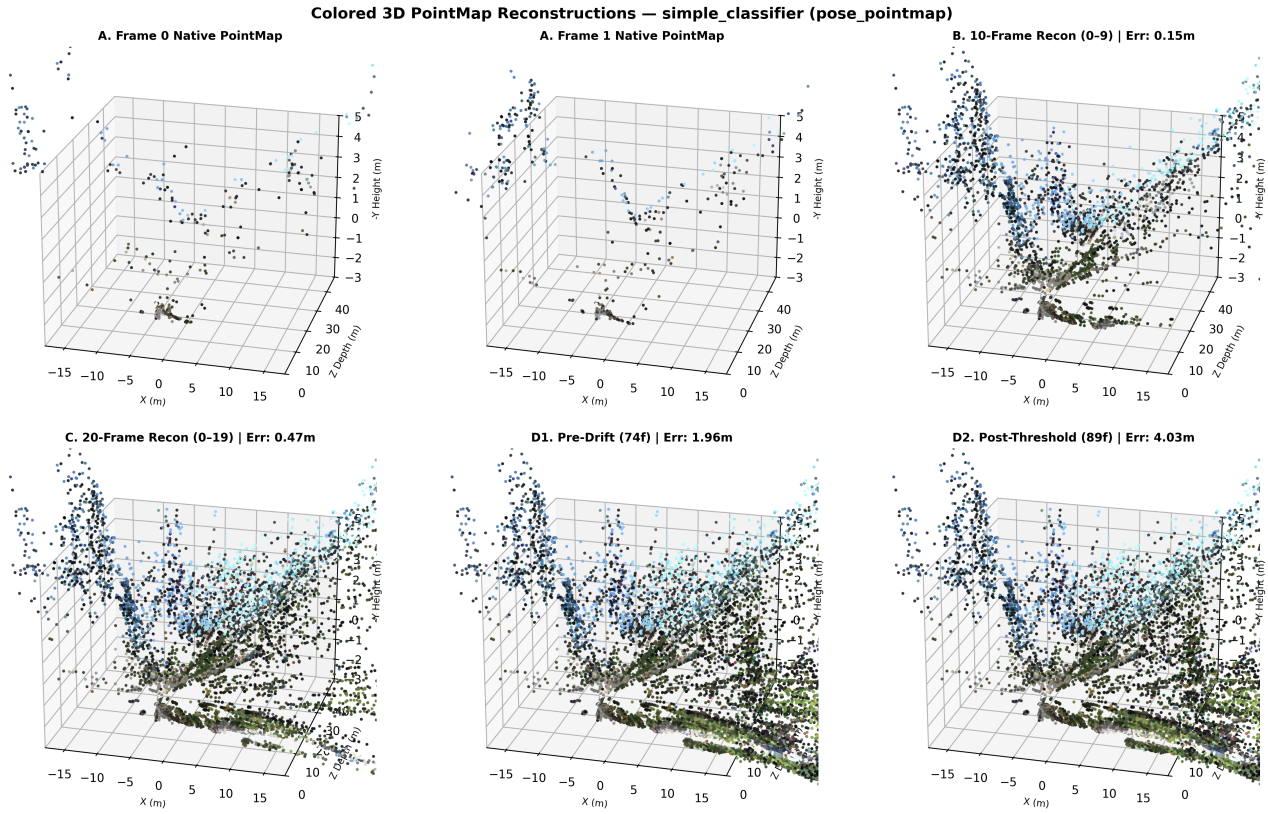



Fig. 7. Colored 3D PointMap reconstructions for the `simple_classifier` (pose–pointmap configuration) on KITTI Sequence 10. (A) Native single-frame point clouds for frames 0 and 1. (B) 10-frame accumulated reconstruction ($t \in [0, 9]$) exhibiting sub-decimeter metric accuracy (Err $= 0.15$ m)[cite: 8]. (C) 20-frame accumulation ($t \in [0, 19]$) retaining sharp road corridor boundaries (Err $= 0.47$ m)[cite: 8]. (D1) Pre-drift boundary accumulation at 74 frames (Err $= 1.96$ m) just prior to threshold crossing[cite: 8]. (D2) Post-threshold accumulation at 89 frames (Err $= 4.03$ m) displaying structural dilation and blurring caused by accumulated pose divergence[cite: 8].

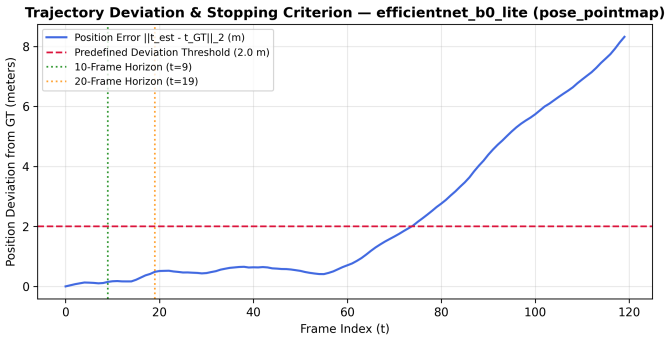



Fig. 8. Trajectory deviation and stopping criterion for `efficientnet_lite0_classifier` (pose–pointmap configuration) on KITTI Sequence 10. The richer spatial context extends the valid reconstruction horizon, delaying the $2.0$ m drift threshold crossing.

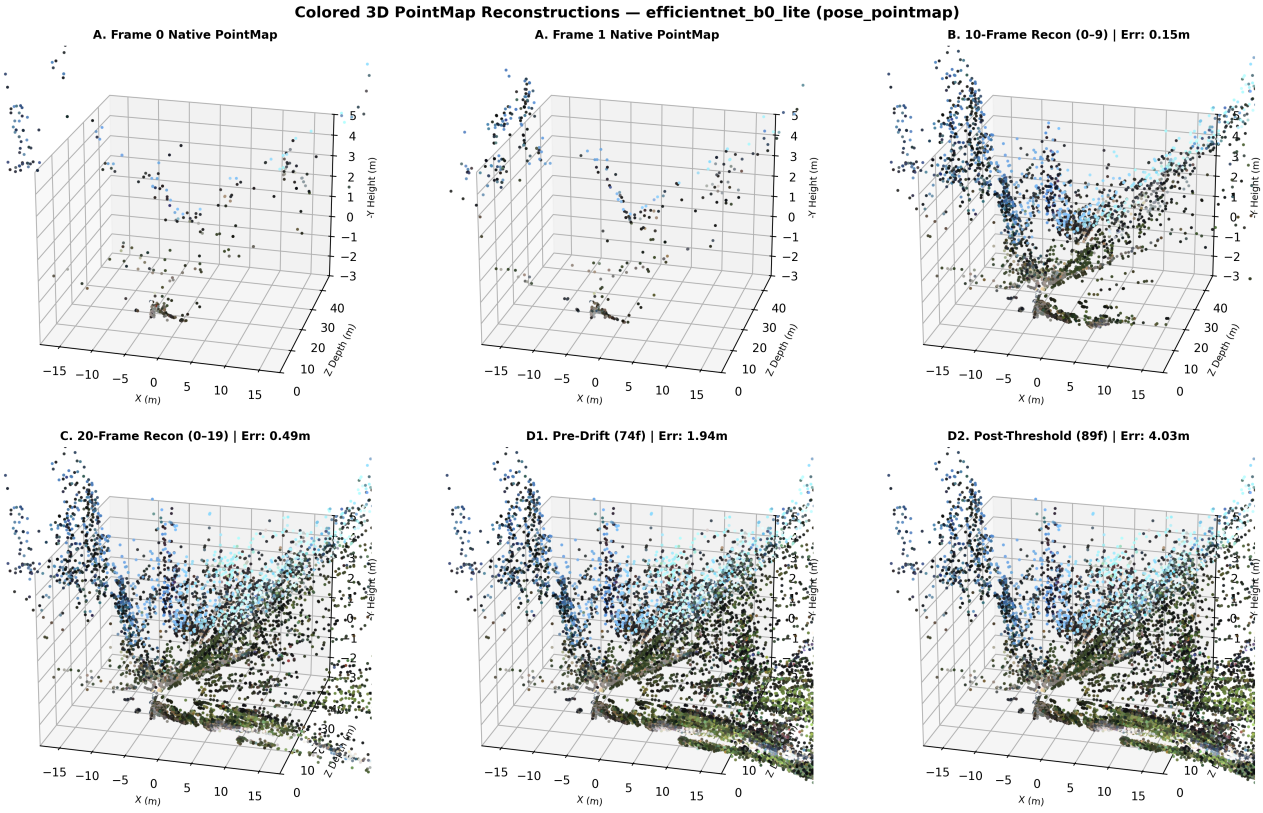



Fig. 9. Colored 3D PointMap reconstructions for the `efficientnet_lite0_classifier` on KITTI Sequence 10. Enhanced epipolar gating prunes dynamic and degenerate edge matches, resulting in sharper structural edges along buildings and reduced point dispersion over extended horizons.